

X-RAY DIFFRACTION ANALYSIS OF PAPER SAMPLES – INVESTIGATION INTO THE EFFECTS OF WATER, DEACIDIFICATION TREATMENTS AND ARTIFICIAL AGEING PROCESS ON CELLULOSE CRYSTALLINITY

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1.0 Introduction

Aqueous conservation treatments of paper materials are traced often by studying changes of paper properties providing mainly macroscopic information e.g. changes in pH, colour, degree of polymerization and mechanical properties.^{1, 2} The aim of this research project was to study the effect of washing, deacidification treatments and artificial ageing process on nanostructural properties of cellulose. X-ray diffraction images of paper samples were obtained in order to determine important nanostructural properties that may reflect macroscopic alterations in paper/cellulose properties. Recent studies with historical paper samples had suggested that crystallization process may be important factor in ageing of cellulose fibres.³

2.0 Methods

Several different analytical methods: XRD, SEM-EDS, CIE $L^*a^*b^*$ colour measurements, pH measurements and IC (Ion chromatography) were used to study paper samples and water based treatment solutions. This paper presents results from X-ray diffraction studies. Other research results are summarized elsewhere.^{4, 5}

2.1 NanoSTAR small angle scattering station technical details

The NanoSTAR facility at Cardiff University, with a sample to detector distance of 4cm, gives wide-angle X-ray diffraction (WAXD) images allowing features of cellulose structures of between 0.22 and 3.0 nm to be observed. A 0.154 nm X-ray beam is generated by a Kristalloflex 760 X-ray generator (Bruker AXS, Germany) and focused using cross-coupled Göbel mirrors and a 3-pinhole collimation system. A HI-STAR 2D detector is used for data collection. The calibration standard was calcite. Details of scattering system and data reduction can be obtained. One analysis of paper by X-ray diffraction is the assessment of paper crystallinity.⁶

Two methods of assessing paper crystallinity, were labelled the crystallinity index (CI) and crystallinity ratio (CR).⁷ These methods use two key values from the data: the height of the (200) reflection, labelled $I_{\max}$, and the height of the scattering at 2.25 nm^{-1} corresponding to diffuse scatter from noncrystalline matter, where scattering is not due to peak maxima, labelled $I_{\min}$. The CI and CR are calculated as:

Crystallinity Index, $CI = (I_{\max} - I_{\min})/I_{\max}$

Crystallinity Ratio, $CR = 1 - I_{\min}/(I_{\max} - I_{\min})$

2.2 Samples

Old “rag” paper samples, which contained mainly pure flax fibres, were taken from the margins of the old book from the year 1831. New paper samples were Whatman no 1, which are pure cotton fibres.

Washing process with Millipore Elix III purified water and deacidification treatments (1 g paper/100ml solution) with 0.02 M $\text{Ca}(\text{OH})_2$ or 0.04 M $\text{Mg}(\text{HCO}_3)_2$ took 30 minutes each. Artificial ageing of paper samples were made in 80°C and 65% RH for two weeks in Binder Climate chamber.

3.0 Results and Discussion

Figure 1 shows the 2D high-angle X-ray diffraction pattern obtained on the NanoSTAR at Cardiff University of a new untreated paper reference sample at a sample to detector distance of 4 cm, allowing dimensions in the region of 10-0.2 nm to be resolved. The 2D image was converted to the 1D linear profile of intensity (I) versus scattering angle, represented as reciprocal nanometres. The main scattering peak of cellulose (200) is highlighted as $I_{\max}$. $I_{\min}$ at 2.25 nm^{-1} is also highlighted as it was required in Crystallinity calculations.

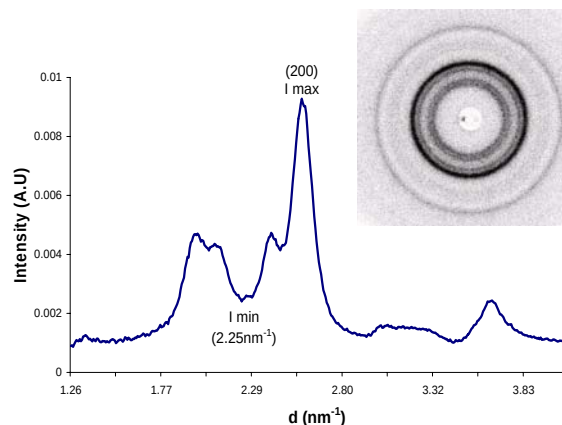


Figure 1. On the top a 2D X-ray diffraction image of a new untreated Whatman paper sample. On the bottom a 1D linear trace of the 2D X-ray diffraction image.

For the studied samples the CI and CR values have been calculated and presented in tables 1– 4.

Old Paper samples	CI	CR
Reference (untreated)	0.660	0.484
Reference (untreated)	0.668	0.504
Washed in deionised H_2O	0.669	0.506
Washed in deionised H_2O	0.672	0.512
Treated with $\text{Ca}(\text{OH})_2$	0.667	0.502
Treated with $\text{Ca}(\text{OH})_2$	0.689	0.549
Treated with $\text{Mg}(\text{HCO}_3)_2$	0.586	0.293
Treated with $\text{Mg}(\text{HCO}_3)_2$	0.651	0.463

Table 1. Crystallinity Index and Crystallinity Ratio of the old paper samples. Crystallinity Index Std dev ± 0.0109 , Crystallinity Ratio Std dev ± 0.0241

New Paper samples	CI	CR
Reference (untreated)	0.725	0.621
Reference (untreated)	0.721	0.612
Treated with $\text{Ca}(\text{OH})_2$	0.732	0.634
Treated with $\text{Ca}(\text{OH})_2$	0.729	0.629
Treated $\text{Mg}(\text{HCO}_3)_2$	0.723	0.618
Treated $\text{Mg}(\text{HCO}_3)_2$	0.720	0.611

Table 2. Crystallinity Index and Crystallinity Ratio of the new paper samples. Crystallinity Index Std dev ± 0.0049 , Crystallinity Ratio Std dev ± 0.0092

Aged Old Paper samples	CI	CR
Reference (aged)	0.721	0.613
Reference (aged)	0.714	0.600
Washed in deionised H_2O	0.671	0.510
Washed in deionised H_2O	0.676	0.522
Treated with $\text{Ca}(\text{OH})_2$	0.671	0.510
Treated with $\text{Ca}(\text{OH})_2$	0.670	0.507
Treated with $\text{Mg}(\text{HCO}_3)_2$	0.704	0.580
Treated with $\text{Mg}(\text{HCO}_3)_2$	0.670	0.507

Table 3. Crystallinity Index and Crystallinity Ratio of the artificially aged old paper samples. Crystallinity Index Std dev ± 0.0220 , Crystallinity Ratio Std dev ± 0.0458 .

Aged New Paper samples	CI	CR
Reference (aged)	0.744	0.655
Reference (aged)	0.742	0.652
Treated with $\text{Ca}(\text{OH})_2$	0.732	0.635
Treated with $\text{Ca}(\text{OH})_2$	0.751	0.668
Treated with $\text{Mg}(\text{HCO}_3)_2$	0.742	0.652
Treated with $\text{Mg}(\text{HCO}_3)_2$	0.735	0.639

Table 4. Crystallinity Index and Crystallinity Ratio of the artificially aged new paper samples. Crystallinity Index Std dev ± 0.0066 , Crystallinity Ratio Std dev ± 0.0120

The conclusions from these data are:

- Ageing process may be inducing a certain amount of Crystallinity.
- Naturally aged rag paper samples have a reduced Crystallinity when compared with the new Whatman paper samples.
- Treatments do not appear to have any strong effect on Crystallinity with the exception of the aged old rag paper samples.
- Washing and deacidifying of rag paper prevent further crystallization process during ageing.

There is one key feature that is of interest in these samples, namely the samples treated with $\text{Ca}(\text{OH})_2$. In these samples it is clear that calcite deposits remain on the paper. The X-ray

diffraction image shows a clear ring of sharp points at the highest angles (See figure 2).

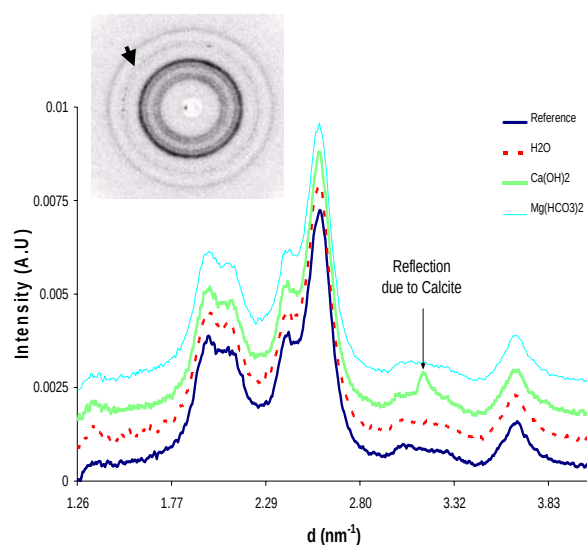


Figure 2. On the top X-ray diffraction image of old paper treated with $\text{Ca}(\text{OH})_2$. Calcite reflection has been labelled with an arrow. 1D linear trace of the 2D X-ray diffraction images for the old paper samples set. Calcite reflection due to treatment with $\text{Ca}(\text{OH})_2$ has been labelled. (Linear traces offset).

The presence of the calcite reflection is most likely a remnant of the processing with $\text{Ca}(\text{OH})_2$, and has been observed in other materials which are subjected to treatment with $\text{Ca}(\text{OH})_2$.⁸

Results show that aqueous conservation treatments, washing and deacidification have the effect on Crystallinity of artificially aged rag paper cellulose. Anyhow, examination with other, more complex papers which contain different cellulose fibres e.g. wood fibres, lignin and additives, still remain the subject of further studies.

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